Real-life application of Regression: Neuro Genetics

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1 What is Regression ?

We start by saying regression is a statistical method where we model and analyse the mathematical relationship between two or more variables. One is the dependent variable, and the others are the independent variables. It estimates the total conditional expectation of the dependent variable given the independent variables, mapping out the change in the target as the predictors shift.

2 What is Neuro Genetics ?

So Neuro Genetics is the study of how an individual's genetic code plays a part in the development, function, and also the disorders of the nervous system. It's the intersection of Neuroscience and Genetics. It examines how specific DNA variations play the role from basic cognitive ability to the risk of developing conditions of Alzheimer, Parkinson's and even Autism!

3 Understanding the Statistical Model

As for us statistics students, Neuro Genetics is simply viewed as a high-dimensional dataset of X and Y variables.

3.1 Defining the Variables

- **Predictor Variable (X):** This is going to be the independent variable which is going to represent the genetic data. In simple statistical terms, we say it is viewed as a discrete numerical variable where, $X \in \{0, 1, 2\}$ that counts how many copies of a specific genetic mutation a person carries.
- **Response Variable (Y):** This is going to be the dependent variable where it indicates the neurological outcome being measured. It is something which can be a continuous variable or discrete variable too, such as

a cognitive memory test score or brain volume, or a binary variable, like 1 if a patient has a condition like Alzheimer's and 0 if they do not.

4 Applying Regression

So depending on the type of our response variable which is Y in this instance, we are gonna apply different regression models to quantify the genetic relationships together.

4.1 Linear Regression for Continuous Traits

When Y is continuous, we use the standard multiple linear regression model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \epsilon$$

Here, β_1 represents the "effect size" of the genetic variant X_1. If the p-value associated with β_1 is statistically significant, we will reject the null hypothesis and conclude that this is the specific genetic variant that impacts the neurological trait. Additional predictors like X_2 (age) and X_3 (gender) are included as covariates to control for unnecessary noise.

4.2 Logistic Regression for Disease Risk

When Y is binary (disease presence vs. absence), standard linear regression is going to fail because probabilities must lie between 0 to 1. So, we will apply the logistic regression to model the probability p of developing the neurological disorder:

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots$$

This equation calculates the log-odds of a neurological condition occurring, allowing us to predict disease risk based directly on a patient's genetic profile.

section Genome-wide association studies (GWAS) GWAS is one of the most common applications of regression in genetics. Instead of looking at one gene at a time, we can scan the entire human genome to find the variations associated with any disease like Autism. So, statistically, it doesn't only run on one regression model. They run millions of millions of independent regression model simultaneously, one for each genetic variant ever recorded.

The GWAS Statistical Challenge

The Multiple Testing Problem

So here's a thing: if we run millions of millions of regression models, it creates a huge statistical problem: If we test 5 million genes using a standard significance level of $\alpha = 0.05$, we would expect 250,000 false positives purely by random

chance!
Bonferroni Correction: The solution we need! We first divide our target "α" by the total number of tests (n).

$$\alpha_{GWAS} = \frac{0.05}{5,000,000} = 5 \times 10^{-8}$$

Only genes with a p-value smaller than 5×10^{-8} are considered statistically significant.

Polygenic Risk Scores (PRS)

Handling Complex Diseases

Many neurological conditions like Autism or schizophrenia are not caused by a single gene. Instead, they are caused by thousands and thousands of different genes, each having a very tiny effect that is not noticeable in a single one.
To counter this, statisticians use **Polygenic Risk Scores (PRS)**. Because we have more predictor variables, which are the genes in this case, than we have patients, then the standard regression is going to fail. We must use **Penalised Regression** (like Lasso or Ridge Regression) to shrink the coefficients and handle highly correlated genetic data.

The Math Behind PRS

Calculating the Risk Score

Once we start running our penalised regression, we are going to get a weight (β) for every single gene. We calculate the total Polygenic Risk Score for a new patient using a simple linear combination that goes as follows:

$$PRS = \sum_{i=1}^{k} \beta_i X_i$$

Where:

- β_i = the regression coefficient (weight) of the i-th gene.
- X_i = the number of copies of that gene the patient has: $0, 1$, or 2.

This is going to give doctors a single number representing the patient's genetic risk for a neurological disorder.

Imaging Genetics

Linking Genes to Brain Scans

There's another real-world major application and that is Imaging Genetics. Here, we take the Y variable, which is taken directly from an MRI scan of the brain, for instance, the volume of the hippocampus, which shrinks in Alzheimer's patients.

Researchers use the **Multivariate Regression** to map genetic data X against these brain images Y. This is going to allow us to mathematically prove that a specific DNA mutation can physically alter the structure of the brain.

Wrapping Up

Why Statistics Matters in Biology

Without having statistics by our side, the human genome is just a massive impenetrable walls of, unreadable string of billions of letters.

where:

- Regression acts as the ultimate filter.
- It isolates the true biological signals from random noises.
- It also allows us to predict diseases, understand the brain structure, and push the boundaries of medicines all by using some foundational statistical equations.